

Lewis Structures

Answer key with worked solutions

A. Counting electrons

1. Valence electrons.

(a) $4 + 4 \times 1 = 8$ (b) $5 + 3 \times 7 = 26$ (c) $5 + 4 \times 1 - 1 = 8$ (d) $4 + 3 \times 6 + 2 = 24$

B. Drawing structures

2. H_2S , NF_3 , CCl_4 .

(a) 8 e^- : $\text{H}-\text{S}-\text{H}$, **2 lone pairs on S** (b) 26 e^- : N bonded to three F, **1 lone pair on N**, 3 lone pairs on each F (c) 32 e^- : C bonded to four Cl, no lone pairs on C, **3 lone pairs on each Cl**

3. CH_2O and C_2H_4 .

(a) 12 e^- : C single-bonded to two H and **double-bonded to O**; 2 lone pairs on O. (b) 12 e^- : **$\text{C}=\text{C}$ double bond**, each C single-bonded to two H; no lone pairs.

4. CS_2 .

16 e^- (like CO_2): **$\text{S}=\text{C}=\text{S}$** , with 2 lone pairs on each S: **4 lone pairs** in total.

5. NH_4^+ .

8 e^- : N bonded to four H, no lone pairs, drawn in square brackets with a + charge. Formal charge on N = $5 - 0 - \frac{1}{2}(8) = +1$ (equal to the ion's charge).

C. Formal charge and resonance

6. Ozone.

$\text{O}=\text{O}-\text{O}$: the central O has 1 lone pair, the double-bonded O has 2 lone pairs and the single-bonded O has 3 lone pairs. Formal charges: double-bonded O $6 - 4 - 2 = 0$; central O $6 - 2 - 3 = +1$; single-bonded O $6 - 6 - 1 = -1$. They add up to 0, the charge of the molecule.

7. Nitrate.

24 e⁻: N in the centre with **one N=O double bond and two N-O single bonds**. The three resonance structures differ only in which O has the double bond; they are linked by ↔ and drawn in brackets with a - charge. Formal charges: N +1, each single-bonded O -1, double-bonded O 0. Because the real ion is a hybrid, **all three N-O bonds are the same length**, between a single and a double bond.

D. Exceptions

8. BF₃.

Boron has only 3 valence electrons and forms three single B-F bonds, so it is surrounded by only **6 electrons**, an incomplete octet.

E. Challenge

9. XeF₄.

Xe is bonded to four F, each F with 3 lone pairs (8 + 24 = 32 e⁻). The remaining 4 e⁻ go on Xe as **2 lone pairs**. Xenon is surrounded by 8 + 4 = **12 electrons**, an expanded octet. This is possible because xenon is a period-5 element, large enough to hold more than 8 electrons in its valence shell.